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THE EFFECTS OF TIDAL FORCES ON THE STABILITY AND DISPERSAL RATES OF A PROTOPLANET

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SUMMARY

The tidal disruption of a protoplanet by the Sun is considered by both analytical and numerical means. It is concluded that the disruption process is very rapid and that this implies that most conventional protoplanetary theories for the formation of planets need to be modified otherwise complete disruption would have occurred before a refractory core could form.

INTRODUCTION

Any known raw material that could have featured in the formation of the planetary system must have had a composition similar to that of the Sun or interstellar material. In order to form the Earth and the other terrestrial planets from such material, segregation of the heavier elements from the hydrogen and the helium must therefore have occurred. For the major planets this process of segregation need not have occurred since they have an overall composition similar to that of the Sun and, in fact, Donnison & Williams (1974) have shown that the main features of Jupiter can be explained by the gravitational contraction of an object of solar composition. There are two obvious alternative hypotheses: segregation of the heavy elements can occur either before, or after, the material has gathered together to form protoplanets.

A number of theories have been proposed in which the segregation occurs prior to the agglomeration into planets. In these theories a disc or ring of material is captured about the Sun. In the regions closest to the Sun only the non-volatile material accumulates, first into grains and eventually into planets. This process, therefore, gives birth to terrestrial type planets. Further away from the Sun, in the cooler parts of the disc, the growing embryos also capture the gas and so develop into major planets. Typical theories which employ variations on this basic theme have been proposed by Cameron (1962), Hoyle (1960), Pendred & Williams (1968), Safranov (1969) and Schatzmann (1967).

In the alternative hypothesis, material is first collected into protoplanets, with the segregation subsequently occurring within these protoplanets. Such theories have been proposed by McCrea (1960), Urey (1966) and Woolfson (1964). If they are to produce terrestrial planets, these theories are in fact faced with two difficulties. First, the heavy elements must be separated from the hydrogen and helium and secondly, the hydrogen and helium, which is not required, must be removed from the protoplanet.

The first problem has been fairly thoroughly investigated. McCrea & Williams

(1965), Williams & Crampin (1971) and Williams & Handbury (1974) have all concluded that the heavy elements, in the form of grains, can settle to the centre of such a protoplanet in a reasonably short period of time, of say 1000 yr. There therefore remains only the problem of removing the outer layers of hydrogen and helium in order to form a terrestrial planet. Whatever mechanism is proposed, it must only be efficient in the neighbourhood of the terrestrial planets since the major planets must not lose their hydrogen and helium. McCrea (1960) suggested that the tidal effect of the Sun was an appropriate mechanism, noting that the Roche limit (roughly the boundary between stability and instability against the tidal effect of the Sun) for an object with his assumed density for a protoplanet coincided with the distance of the asteroidal belt. Hence the terrestrial protoplanets would have suffered disruption while the remainder would not.

It is this tidal dispersal process that we wish to discuss in this paper. The situation is not as simple as envisaged by McCrea, for there is a need to form the heavy element core in the protoplanets before dispersal takes place. It is, therefore, of vital importance to know how rapidly tidal dispersal occurs. If it is a slow process, then there may be time for a heavy element core to form before dispersal occurs on an appreciable scale. Therefore we initially estimate, in a fairly simple minded way, the dispersal time for a protoplanet moving on a circular orbit in the terrestrial planet region. If this time turns out to be short, then the protoplanet must have been moving in a different fashion when segregation occurred. Since all the theories that predict the existence of protoplanets also suggest that they would be formed on highly eccentric orbits it does not seem unreasonable to postulate that the protoplanets moved on eccentric orbits. It has been suggested that the resisting effect of residual gases was responsible for rounding the orbits to the present-day near circular shape. The investigation of such resistive effects is beyond the scope of this paper, however, the main effect is obvious. The resisting medium moves on Keplerian circular orbits, hence it slows down the planet at perihelion and speeds it up at aphelion. In consequence the perihelion distance is not changed considerably in the process, and is certainly not decreased. We therefore investigate the dispersal of a protoplanet moving on an elliptic orbit with its perihelion in the region of the present day terrestrial planets. Such a protoplanet will spend the major part of its life in a region where it is tidally stable and will only enter the region of instability near perihelion.

Such a protoplanet could remain stable while the core formed by segregation and would only become unstable when resisting effects had rounded off the orbit until it became a near circular orbit almost wholly within the region of instability. Such an investigation will be carried out numerically though a simple analytical approach will also be used. With such a programme it is natural to reinvestigate the limiting case of circular orbits and so obtain a check that the simple analytical solution and the numerical solution agree with each other. The numerical technique involves calculating the orbit of a small particle detached from the main protoplanet and we shall follow the motion of this particle in an attempt to determine where disrupted material finally ends up.

THE TIDAL DISPERSAL OF A CONDENSATION

The shape of a condensation was discussed at length by Jeans (1917, 1928) and Darwin (1906) early this century. However, they made no attempt to obtain the

time taken for dispersal and since this is the parameter of interest to us, we shall approach the problem slightly differently.

In reality the shape of the protoplanet will become distorted as it moves inside the Roche limit and the gravitational field generated by it will differ from that of a sphere. It becomes impossible to follow the motion however if we attempt to take account of this changing shape. Any tendency towards having a central condensation within the protoplanet brings its gravitational field nearer to that of a point mass and since, even in the limiting shape of the body given by Jeans (1917) the difference between the two fields is small at reasonable distances from its centre, we believe that a reasonable approximation is given by taking the field of the protoplanet to be that of a point mass, M , placed at its centre. The orbit of this point mass is assumed to be an ellipse of given eccentricity e and perihelion q (or semi-major axis (a)) about the Sun at one focus, the Sun also being taken to be a point mass $M_{\odot}$.

What we attempt to do is to calculate the orbit of a small third mass, taken to be just detached from the protoplanet. If this third object moves away from the protoplanet, then disruption is taking place while the protoplanet will be stable if this third object wants to move towards the centre of the protoplanet. The problem being tackled is therefore the classical three-body problem, or at least one special case of it. Other special cases have, of course, been studied many times before.

For the case of the circular orbit for the protoplanet, all that is needed, as far as the numerical results are concerned, is to place $e = 0$ in the general program. However, an analytical approach is also possible for this case and we shall discuss this approach first.

DISRUPTION OF A PROTOPLANET ON A CIRCULAR ORBIT

In this section we wish to investigate the disruption of a protoplanet, moving on a circular orbit about the Sun, by analytical means. If the radius of the orbit is R , then the angular velocity with which the protoplanet describes its orbit, ω_1 , is given by

$$GM_{\odot} = R^3\omega_1^2. \quad (1)$$

Consider now the innermost particle of this protoplanet assumed to be free. It is at a distance r_0 nearer to the Sun. If it is to be in equilibrium, then it must move with an angular velocity ω_2 given by

$$(R-r_0)\omega_2^2 = \frac{GM_{\odot}}{(R-r_0)^2} - \frac{GM}{r_0^2}. \quad (2)$$

This equation is not, of course, the general equation for any particle on the surface for then internal pressure terms should also be included. It is, however, applicable when the disruption is imminent for then the pressure forces will have become zero.

After an interval of time τ , the angular separation between the radius vectors from the Sun to the planet and to the particle, θ say, is given by

$$\theta = \tau|\omega_2 - \omega_1|, \quad (3)$$

or

$$\tau = \frac{\theta}{|\omega_2 - \omega_1|}. \quad (4)$$

Substituting from equations (1) and (2) for ω_1 and ω_2 gives

$$\tau = \left| \frac{(R-r_0)^{3/2} R^{3/2} r_0 \{R^{3/2} [r_0^2 M_\odot - (R-r_0)^2 M]^{1/2} + r_0 (R-r_0)^{3/2} M_\odot^{1/2}\}}{R^3 [r_0^2 M_\odot - (R-r_0)^2 M] - r_0^2 (R-r_0)^3 M_\odot} \right| \theta / \sqrt{G}. \quad (5)$$

For most cases of interest $R \gg r_0$ and in this event equation (5) simplifies to give

$$\tau = \theta / \sqrt{G} \left| \frac{R r_0 \{R^{3/2} [r_0^2 M_\odot - R^2 M]^{1/2} + r_0 R^{3/2} M_\odot^{1/2}\}}{3 r_0^3 M_\odot - M R^3} \right|. \quad (6)$$

This equation gives the time required for a displacement θ to occur between the radius vectors from the Sun to the protoplanet and particle respectively. In order to estimate the dispersal time, we must select a value for θ which is fairly small and at the same time has a physical meaning which can be reproduced in the numerical discussion of the same problem. Such a value is obviously θ_0/R .

From inspection of equation (6), it is obvious that the dispersal time tends to infinity as

$$M R^3 \rightarrow 3 r_0^3 M_\odot.$$

In other words, no dispersal can occur if $R = R_c$ where

$$R_c = \left[\frac{3 M_\odot r_0^3}{M} \right]^{1/3} = 0.8947 \left[\frac{M_\odot}{\rho} \right]^{1/3} \quad (7)$$

where $\rho = 3M/4\pi r_0^3$ is the mean density of the protoplanet. R_c corresponds in our model to the Roche limit given by Darwin (1906) as

$$R_c = 1.523 \left[\frac{M_\odot}{\rho} \right]^{1/3}. \quad (8)$$

It is interesting to note that if we compensate for a change in density between the two models because the Darwin model assigns the correct ellipsoidal shape, and consequent smaller volume, to the protoplanet, then the two expressions for R_c are in almost complete agreement.

If the protoplanet is to be disrupted then its distance from the Sun must be

$$R = x R_c, \quad (9)$$

with $x \leq 1$.

In this event

$$r_0^2 M_\odot - R^2 M = r_0^2 M_\odot (1 - 3x^2 r_0/R_c),$$

while $3r_0^3 M_\odot - R^3 M = 3r_0^3 M_\odot (1 - x^3)$.

Equation (6) therefore gives the dispersal time as

$$\begin{aligned} \tau &= \frac{1}{3} \sqrt{\frac{R_c^3}{GM_\odot}} \left(\frac{x^{3/2}}{1-x^3} \right) \left\{ 1 + \left(1 - \frac{3x^2 r_0}{R_c} \right)^{1/2} \right\} \\ &\doteq \frac{P}{3\pi} \cdot \frac{x^{3/2}}{1-x^3}, \end{aligned} \quad (10)$$

where P is the period of an orbit of radius R_c . Equation (10) thus shows that unless x is close to unity, so that the orbit is close to the Roche limit, dispersal occurs very rapidly in comparison with the orbital time.

TABLE I

Distance (cm)	Planet	Dispersal time (yr)	
		Analytic	Numerical
6×10^{12}	Mercury	0.011	0.0128
10^{13}	Venus	0.037	0.0364
1.5×10^{13}	Earth	0.0826	0.0742
2.5×10^{13}	Mars	0.23	0.181
4×10^{13}	Asteroids	1.49	0.593
8×10^{13}	Jupiter	Stable	Stable

In our numerical discussion, to be given later, we will, however, consider specific models. To illustrate further the rapidity of the dispersal process we shall, therefore, calculate the dispersal time, using equation (6). We will, therefore, consider the numerical values for a protoplanet basically similar to the type considered by McCrea (1960) and for convenience consider an object with a mass of $10^{-3} M_{\odot}$ and a radius of some 3×10^{12} cm. For such an object, in round numbers

$$R_c = 4 \times 10^{13} \text{ cm},$$

hence the protoplanets at terrestrial distances would have been within the critical distance of the Sun. The relevant dispersal times for various protoplanetary distances are shown in Table I. These values again bear out our contention that the dispersal time is rapid unless the object is very close to its Roche limit.

DISRUPTION OF A PROTOPLANET ON AN ELLIPTIC ORBIT

It is obviously a much more difficult task to calculate the dispersal time of an object moving on an elliptic orbit, since this dispersal time will be a function of the relative position on the orbit. However, we may calculate the dispersal time at perihelion in the following way.

At perihelion, the radius of curvature for an ellipse is given by

$$\rho = a(1 - e^2)$$

a being the semimajor axis and e the eccentricity.

Hence the equation for the protoplanet corresponding to equation (1) becomes

$$\frac{V_1^2}{a(1 - e^2)} = GM_{\odot}/a^2(1 - e)^2, \quad (11)$$

with $V_1 = a(1 - e) \omega_1$. Hence $\omega_1^2 = GM_{\odot}(1 + e)/a^3(1 - e)^3$. For the innermost particle the equation corresponding to equation (2) is

$$\frac{V_2^2}{[a(1 - e^2) - r_0]} = GM_{\odot}/[a(1 - e) - r_0]^2 - GM/r_0^2 \quad (12)$$

with $V_2 = [a(1 - e) - r_0] \omega_2$. Performing similar algebraic manipulations to those carried out in Section 1 gives

$$\tau = \frac{2\theta}{\sqrt{GM_{\odot}}} [a(1 - e)]^{5/2} r_0^2 [r_0^3(3 + 4e) - a^3(1 + e)(1 - e)^3 M/M_{\odot}]^{-1} \quad (13)$$

as the dispersal time.

Again, defining $R_c^3 = 3r_0^3 M_{\odot}/M$, and letting perihelion distance be zR_c , with

$\theta = r_0/zR_c$, equation (13) becomes

$$\tau = (1+e)^{1/2}(1-e)^{3/2} \frac{P}{\pi} ((3+4e) - 3(1+e)z^3)^{-1} \quad (14)$$

where P is the period of the elliptic orbit. We may also easily calculate the time for which the protoplanet with perihelion distance zR_c is within a distance R_c of the Sun, using Kepler's law of areas. Performing the necessary algebra, this turns out to be

$$\tau_1 = \frac{P}{\pi} (1-e^2)^{3/2} I, \quad (15)$$

where I is a dimensionless integral which arises from the evaluation of the area swept out by the radius vector from the Sun to the protoplanet. The important feature of I is that $I \rightarrow 0$ only if $z \rightarrow 1$. If we require the protoplanet to be stable, then clearly we need $\tau_1 < \tau$ and it is immediately evident that this can only happen if $z \rightarrow 1$. Thus analytically we predict that a protoplanet moving on an elliptic orbit can be stable only if the perihelion distance is close to the critical distance. We shall now investigate the same problem by numerical techniques.

EQUATION OF MOTION FOR A NUMERICAL SOLUTION

As a simplification to the general problem we have to consider the situation where a particle moves under the influence of a stationary Sun and a protoplanet moving on a given ellipse about the Sun. First, we must have a means for determining the position of the protoplanet as a function of time. This can be done either by integrating the equation of motion numerically or by the summing of a series expansion in terms of Bessel functions. We have selected the latter method and from now on we may regard the position (R, θ) motion (functions of $\dot{R}$ and $\dot{\Omega}$) and acceleration (function involving $\ddot{\Omega}$ and $\dot{\Omega}$) of the protoplanet as numerically known at any time.

We now have to determine the equation of motion of the third small particle in the field of the Sun and the protoplanet. It is obviously most convenient to use a rotating frame of reference moving with the protoplanet having the x axis along the radius vector from the Sun to the protoplanet and the y axis coplanar but perpendicular to this, the origin being at the centre of the protoplanet. This is then a rotating set whose rotational velocity Ω is given numerically by the preceding procedure, Ω being perpendicular to the x and y axis.

Using the theory of rotating axes, the acceleration of the particle in an inertial frame is given by

$$\frac{\partial^2 \mathbf{r}}{\partial t^2} + \frac{\partial \Omega}{\partial t} \times \mathbf{r} + 2 \left(\Omega \times \frac{\partial \mathbf{r}}{\partial t} \right) + \Omega \times (\Omega \times \mathbf{r}),$$

where $\mathbf{r} = (x+R)\mathbf{i} + y\mathbf{j}$. This written in component form gives

$$\mathbf{a}_1 = (\ddot{x} + \ddot{R})\mathbf{i} + \ddot{y}\mathbf{j} + \dot{\Omega}(x+R)\mathbf{j} - \dot{\Omega}y\mathbf{i} + 2\Omega(\dot{x} + \dot{R})\mathbf{j} - 2\Omega\dot{y}\mathbf{i} - \Omega^2(x+R)\mathbf{i} - \Omega^2y\mathbf{j}. \quad (16)$$

The force per unit mass on the particle due to the Sun is given by

$$\frac{-GM_\odot}{[y^2 + (R+x)^2]^{3/2}} \{(R+x)\mathbf{i} + y\mathbf{j}\} \quad (17)$$

while the force per unit mass due to the protoplanet is

$$-\frac{GM}{(x^2 + y^2)^{3/2}} \{x\mathbf{i} + y\mathbf{j}\}. \quad (18)$$

Combining equations (16), (17) and (18) gives the equation of motion as

$$\ddot{x} = -\ddot{R} + \dot{\Omega}y + 2\Omega\dot{y} + \Omega^2(x + R) - \frac{GMx}{(x^2 + y^2)^{3/2}} - \frac{GM_{\odot}(R + x)}{\{(x + R)^2 + y^2\}^{3/2}} \quad (19)$$

and

$$\ddot{y} = -\dot{\Omega}(x + R) - 2\Omega(\dot{x} + \dot{R}) + \Omega^2y - \frac{GMy}{(x^2 + y^2)^{3/2}} - \frac{GM_{\odot}y}{\{(x + R)^2 + y^2\}^{3/2}} \quad (20)$$

and these are to be solved given that R , $\dot{R}$, $\ddot{R}$, Ω and $\dot{\Omega}$ have been determined by the above procedure involving Bessel functions. In general these equations will obviously have to be solved by numerical means, and in the actual computations, non-dimensional variables were used defined by $\Lambda = a/A$, $X = x/A$, $Y = y/A$, $Q = M/M_{\odot}$ and $\tau = t/T$, A being a typical scale length and $T = \sqrt{A^3/GM_{\odot}}$.

PROCEDURE FOR DETERMINING THE MOTION

Basically, the above equations were solved numerically using the Runge-Kutta fourth order step-by-step techniques, the step length being adjusted until an acceptable level of accuracy was achieved.

As already mentioned the protoplanet was placed initially at aphelion and the test particle placed on the surface at the point nearest to the Sun as it is a trivial consequence of the analytical work that it is the particle at this point that becomes disrupted first.

Our equations do not contain pressure terms and so they cannot deal with the general motion of the test particle since pressure becomes important if the test particle wishes to move closer to the centre of the protoplanet. If the particle moves further away from the centre than its initial position, then disruption is occurring and the pressure terms become unimportant and our equations valid. Hence, if the test particle moves inside the initial radius of the protoplanet the run is discontinued and a new test particle, on the surface at the nearest point to the Sun is considered. In reality this means that starting from aphelion the test particle appears to stay in the same place relative to the protoplanet until disruption occurs, whereupon it follows an independent path. We calculate this path and if it does not intersect the surface of the protoplanet, then total disruption occurs. If this path does intersect the surface again, the test particle is recaptured and in this event the protoplanet is not disrupted though it loses matter whilst inside the critical distance, which is recaptured when the planet reaches a region of stability. We shall attempt to verify the analytical conclusion that such recapture can only occur if the perihelion distance is close to the critical distance. We shall also compare the estimate of the dispersal time obtained by analytical means with the dispersal time for similar situations obtained numerically.

For any run we have to specify (in non-dimensional values) the eccentricity and semimajor axis of the ellipse on which the protoplanet moves, the mass ratio of protoplanet to Sun and the radius of the protoplanet. Since we are primarily concerned with theories for the formation of terrestrial planets from a protoplanet,

then the dimensions of the protoplanet are almost self-selecting, namely 10^{-3} for Q and a radius of 10^{-2} , and so the only two really free parameters are the eccentricity and semimajor axis. In fact, as it is a more convenient parameter we have used the perihelion distance rather than the semimajor axis as the second parameter. Actual values chosen will be mentioned in the discussion of the results. There are two trivial tests we can carry out on the correctness of the program. First, if the parameters of the ellipse are selected such that the aphelion is within the critical distance, then disruption should start immediately. Secondly, if the perihelion distance is greater than the critical distance, there should never be any disruption. Both these tests were successfully carried out on the program.

DISCUSSION OF THE RESULTS

Calculations have been carried out for a large range of orbital parameters, with e ranging from 0.1 to 0.9 and perihelion covering the whole range of interest. It is not particularly instructive, and space certainly will not permit it, for us to reproduce all these results. Instead, we will display a few results in reasonable detail and summarize only the main points of the whole set of results in Fig. 1.

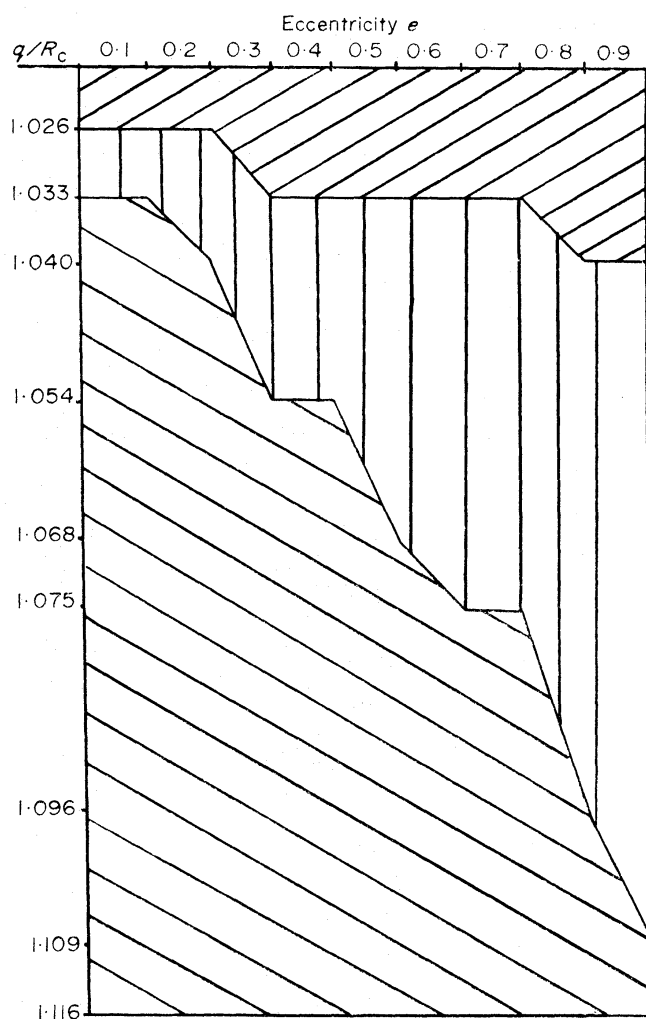


FIG. 1. Plot of e against q/R_c showing the region where disruption (//////), recapture (|||||) and stability (\\\\\\\\) occur, q being the perihelion distance of the orbit.

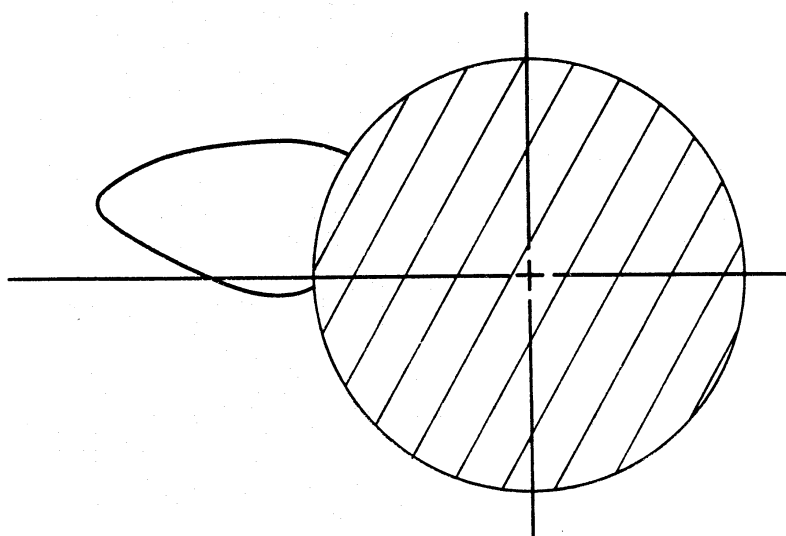


FIG. 2. Motion of a test particle relative to the protoplanet when recapture occurs. The protoplanet is shown as the cross-hatched area.

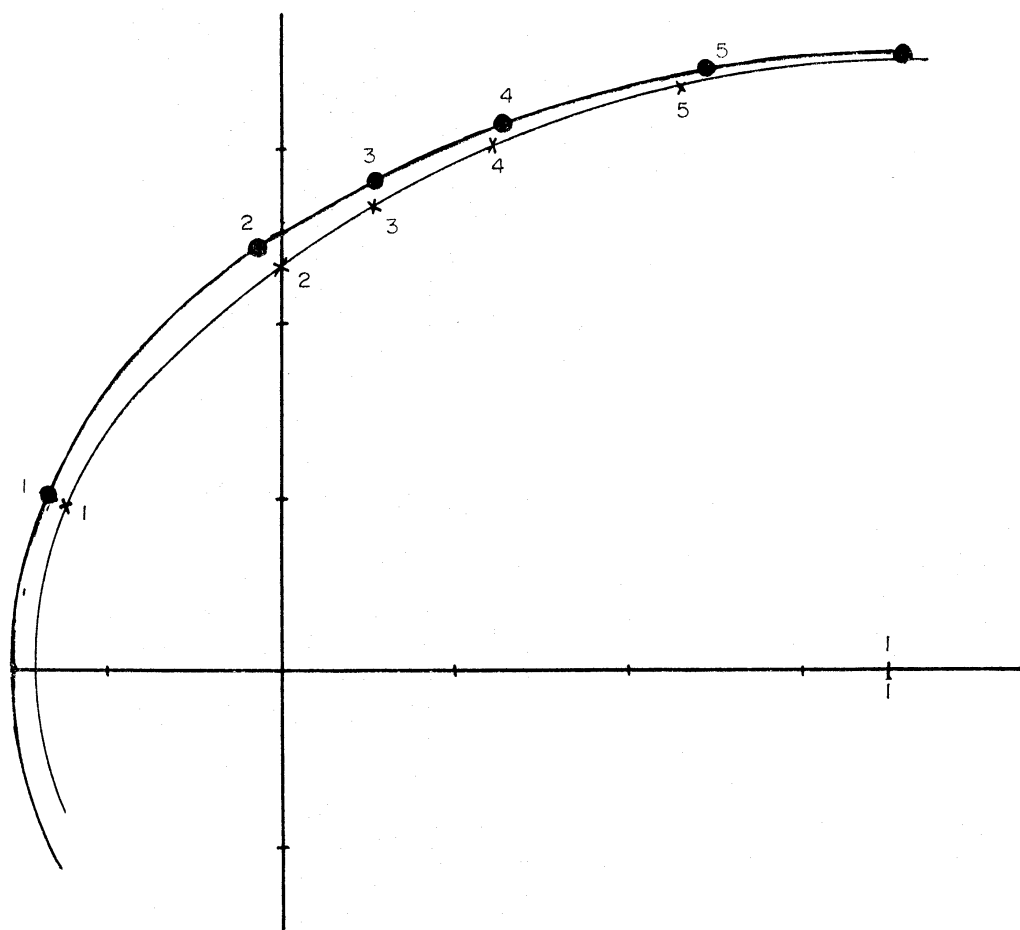


FIG. 3. Motion of a test particle relative to the Sun (at the origin of the axes) when recapture occurs. The outermost curve is part of the orbit of the protoplanet, the labelled filled circles indicating the size and position of the protoplanet at specified times. The corresponding crosses represent the test particle at the same times.

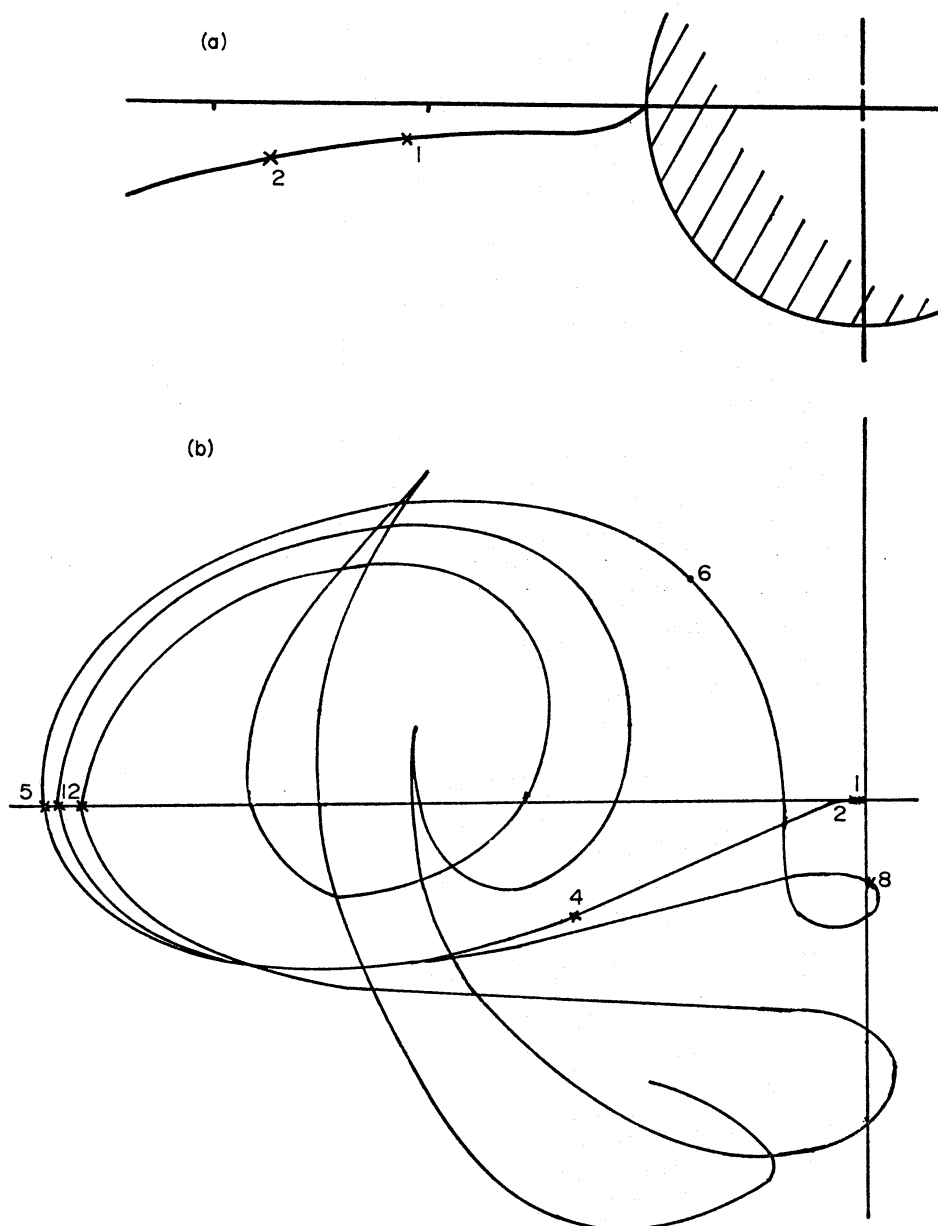


FIG. 4. (a) Motion of a test particle relative to the protoplanet when recapture does not occur (again the protoplanet is the cross-hatched area) to the same scale as Fig. 2. (b) Same event as depicted in Fig. (a) using a scale 25 times smaller, enabling more of the path of the particles to be shown. Numbered crosses again indicate the position of the test particle at specified times.

To display the detailed results, we have selected a value of 0.7 for e . This was selected because at this value, the orbit can clearly be seen to be elliptical while at the same time the major axis is not so elongated as to make the inclusion of the whole orbit in one figure impossible. In Figs 2 and 3 we display the motion of the test particle when the perihelion of the orbit of the protostar is 0.15. In Fig. 2 the motion is shown in the rotating axis, that is the motion relative to the protoplanet, while in Fig. 3 the motion is shown in the inertial reference frame, that is relative to the Sun. Both figures show clearly that the test particle becomes detached after the protoplanet has entered within the critical distance and that it is recaptured

again after the protoplanet has left the critical region. It should be noticed that the critical distance for the protoplanet under consideration is 0.1565. Hence these numerical results verify our analytical conclusion that recapture occurs if the perihelion distance of the orbit is only slightly less than the critical distance.

In Figs 4(a), (b) and 5 we show the results for an orbit of perihelion 0.149, only slightly less than the previous case; again showing the motion of the test particle relative to the protoplanet and Sun respectively. For this case recapture does not occur and so this shows, as we concluded analytically, that recapture can occur only if the perihelion distance is very close to the critical distance.

In Fig. 1 we indicate the region for which recapture (R_c) occurs for a variety of different values of e in terms of the ratio of perihelion distance to critical distance. This just confirms that the conclusions reached above are generally true and have not arisen due to some chance coincidence when $e = 0.7$.

From Figs 4(b) and (5) we can also obtain an indication of where the disrupted material eventually moves to. It appears that it does not either fall into the Sun or escape to infinity. Instead, it follows an orbit not very dissimilar from the protoplanet, differing most at aphelion, but being out of phase with the protoplanet. Though we cannot say definitely that recapture does not occur after very many orbits, it seems to us unlikely since the only region where the two orbits are reasonably close (but out of phase) is close to perihelion, which is within the critical distance and so even if the two objects were momentarily coincident, recapture would not occur.

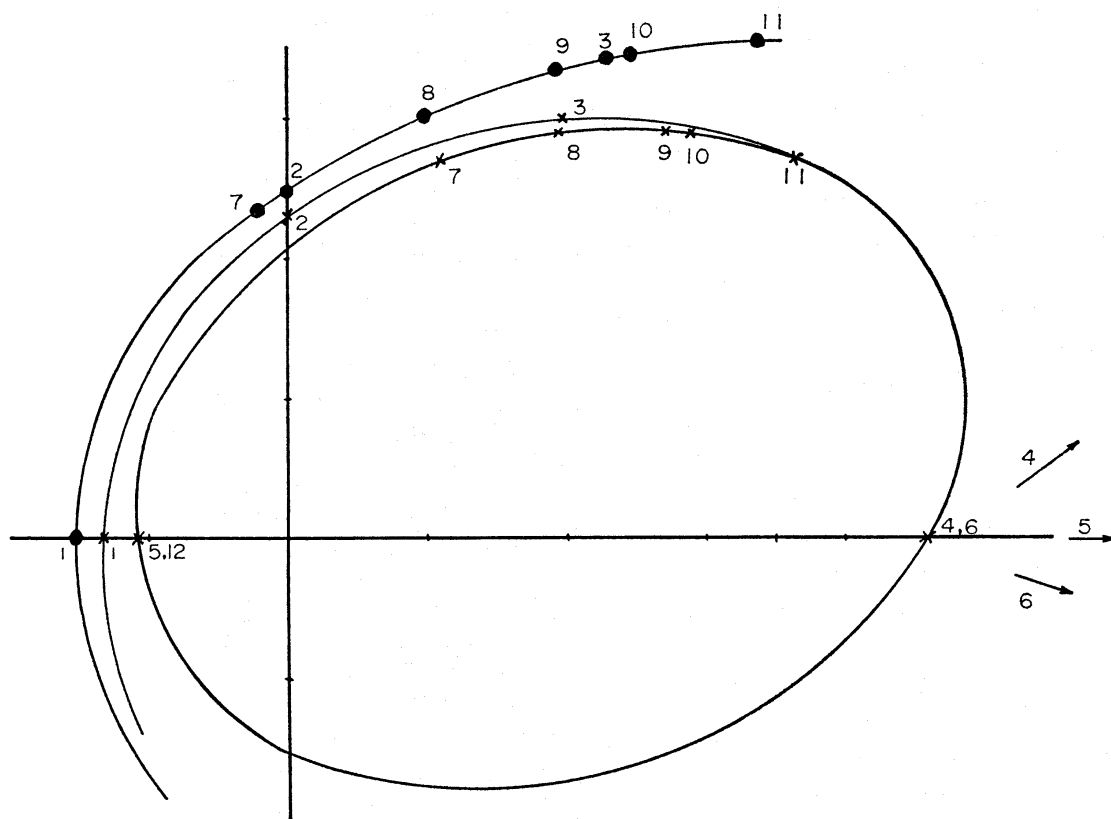


FIG. 5. Motion of a test particle relative to the Sun (at the origin of the axes) when no recapture occurs. The outermost curve is part of the orbit of the protoplanet, the labelled filled circles indicating the size and position of the protoplanet at specified times. The corresponding crosses represent the test particle at the same times.

This aspect of the work is similar to work carried out by Dormand & Woolfson (1971) when investigating the capture of material from a passing star by the Sun, though of course, their star moved on a parabolic rather than elliptic orbit. Our investigations verify their conclusion that the matter is captured in orbit and does not fall into the Sun.

Another feature of interest is the time taken for dispersal to occur. For comparison with the analytical results, we ran the program with $e = 0$, that is circular orbits. For fair comparison, dispersal is deemed to have occurred when the test particle has moved out to a distance $\sqrt{2}$ times the protoplanetary radius, this being the nearest equivalent to an angular separation of r_0/R used analytically. The results of the computations are also shown in Table I, having converted from non-dimensional units to years in the usual way. For comparison the analytical estimates are also included in this table.

Obviously, disruption is a very rapid process and the analytical formula, though very approximate, appears to give very reasonable estimates for the dispersal times.

CONCLUSIONS

The general conclusions regarding the above work are all fairly obvious.

1. If a protoplanet is in a situation where it is subject to disruption, then this disruption occurs quite rapidly.
2. For protoplanets moving on elliptical orbits, it is possible that the perihelion of the orbit is within the critical distance but disruption does not occur.
3. In order that disruption does not occur, the perihelion distance must be very close to the critical distance.
4. The tidal disruptive effect does not remove material from the system, but rather it spreads it out on a roughly similar orbit to that of the original protoplanet.

What is of more interest to us is the implication of these conclusions for the protoplanetary type theories for the origin of the planets.

These theories envisage all four of the terrestrial planets forming from similar protoplanets but it is unlikely that all four such protoplanets could move on orbits with their perihelion close to the critical distance. Since the segregation process investigated by the authors cited earlier takes considerably longer time than that required for disruption, we must conclude that the protoplanetary theory for planetary formation is only feasible provided the protoplanets are stable against the tidal effects of the Sun.

As already mentioned, the critical distance in the theory of McCrea (1960) coincides with the asteroidal distance. It therefore follows from our work that the terrestrial protoplanets envisaged in this theory are unstable and cannot have existed.

Woolfson (1964) does not have any specific protoplanetary densities but if it is related to the density of his protostar, the same difficulty lies in his theory. The protoplanetary density used by Urey (1966) is higher so that no trouble occurs over the Earth and Mars. However, the protoplanetary Venus and Mercury would still have been subject to disruption.

If any protoplanetary theory wishes to explain the existence of even the two major terrestrial planets, then the density of the protoplanet must be of the order of $10^{-6} \text{ g cm}^{-3}$. None of the present theories have densities as high as this.

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